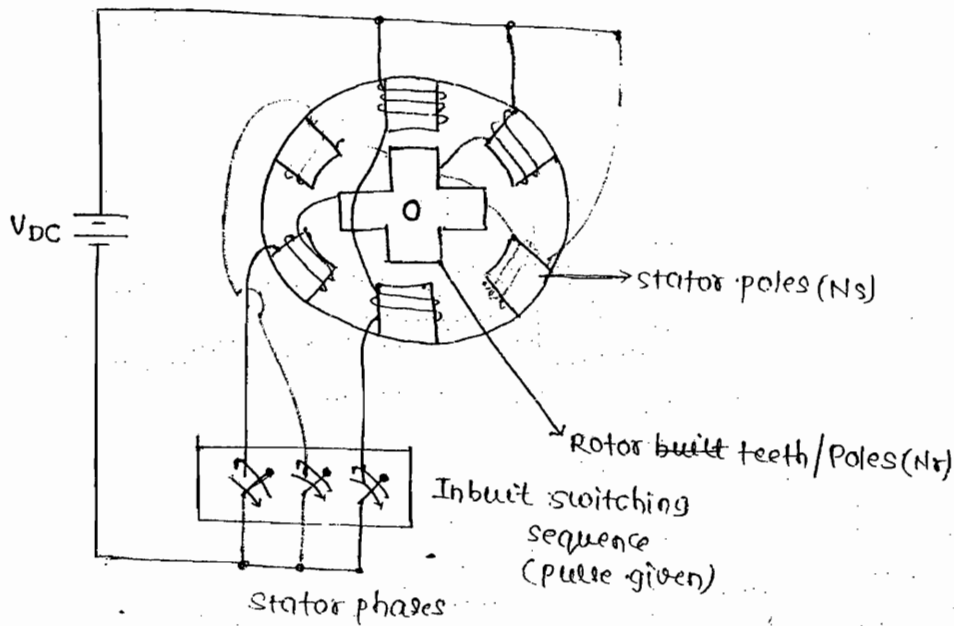


* stepper motor →



$$\text{Step angle } \alpha = \frac{N_s - N_r}{N_s \cdot N_r} \times 360$$

(OR)

$$\alpha = \frac{360}{m \times N_r}$$

- * This are not meant for continuous rotation but rotor rotates in steps by reacting to the pulse given by drive ckt acc to switching seq. by a step angle α in mech. degree.
- * It can be designed depending on no. of poles & teeth in the stator & rotor
- * They can be permanent magnet (or) variable reluctance type.
- * If step freq. is high (pulses are high or microstepping) then the rotor becomes uncontrollable due to its inertia which is called as slew range.

Small Ranges → less than 1w to few watts

Available & popular

Timing devices (clocks.)

Printers, type writer, fax

Computer controls

Robotics

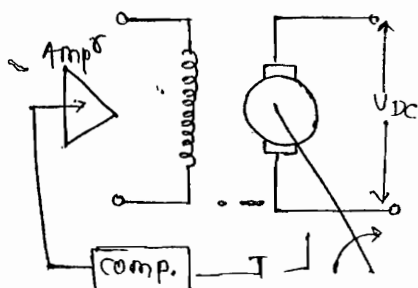
* AC series motor/Universal →

- * If 1ϕ AC supply is applied across dc series motor it continues to run at the torque +ve in both half cycle but the torque is pulsating & not a uniform torque
- * However directly dc series motors are not used but specially designed.
 - (1) stator & rotor contains high grade si-steel laminations to reduce iron loss.
 - (2) Field wdg turns will be comparatively reduced.
 - (3) Arm. reaction, particularly commutation is not successful because of high reactance in the arm. coils.
 - (4) Sparking at the brushes will produce high maintenance repairs which need to be taken care with ~~cu~~ compensated wdg for large ratings.
- * These motors are popular specially less than 1kw ratings because of their high speed (5000-20000 rpm) & high torque within compact size.
- * Due to pulsation in torque they make more noise.
- * Exclusively preferred for food processing like mixers/juicers, vacuum cleaners, portable hand tools like drilling m/c, wood saw's/sewing m/c, recent automatic washing m/c.
- * AC series motors are of high rating around 500HP are used for some traction purposes but with reduced grid frequency are 25Hz only.

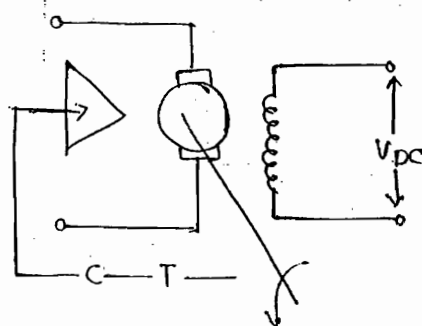
* Servo Motors →

Servo mechanism → Automatic control sys.

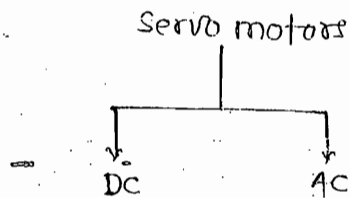
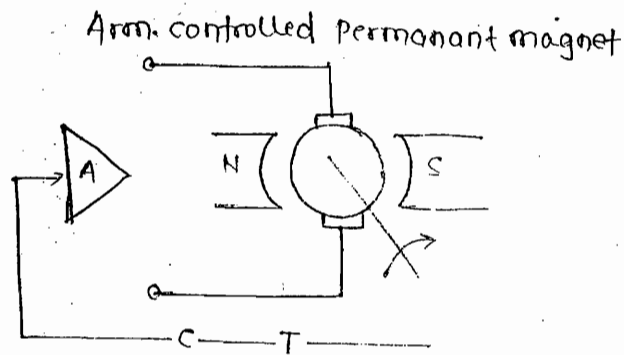
Field controlled DC SM



Arm. cont. dc SM.



- * Compared to field control arm. control servomechanism is fast acting.
- * Arm. control permanent magnet is most popular dc servo motors due to its simple compact design, efficient & no requirement of field wdg & excitation.



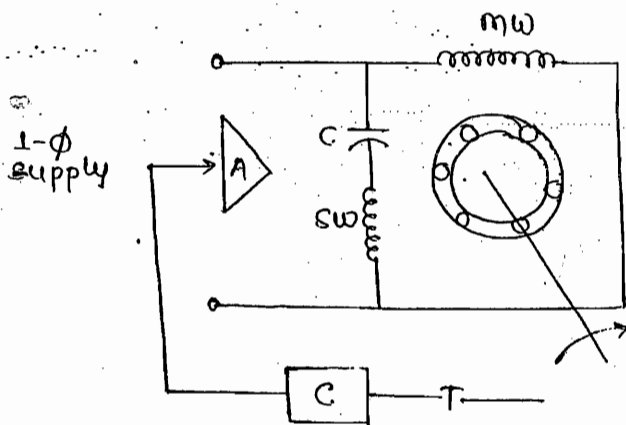
Application →

Control motor used in control sys. apps, radars, process control.
 Position controlling, tracking systems.

AC servo motors →

2- ϕ IM

1 ϕ IM (Capacitor Run type)

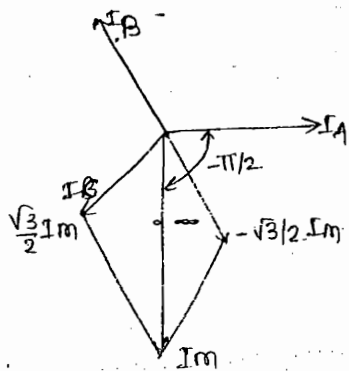


* AC servo motors are basically 1 ϕ IM capacitor run type.

* For quick response their rotor diameters are small & more axial length.

* Compared to conventional there X/R ratio of rotor is low. to get linear T-S c/s for effective response.

16
49



22
57

$$\frac{R_2}{2(2-s)}$$
$$S = \frac{1500 - 1425}{1500}$$

$$S = 0.05$$